

The project is developed through a structured workflow consisting of multiple epics, covering data collection, analysis, preprocessing, model development, and deployment. Each epic focuses on a specific stage of the machine learning lifecycle to ensure systematic and efficient project execution.

****Epic 1: Data Collection and Architecture Design****

* **Story 1:** Download the required dataset from the provided source and store it in the project directory.

* **Story 2:** Define the application architecture, including data flow, machine learning workflow, and deployment structure.

****Epic 2: Visualizing and Analyzing the Data****

* **Story 1:** Import and read the dataset using Pandas for further analysis.

* **Story 2:** Perform univariate analysis to understand the distribution of individual features.

* **Story 3:** Conduct bivariate analysis to identify relationships between two variables.

* **Story 4:** Perform multivariate analysis to uncover patterns involving multiple features.

****Epic 3: Data Pre-Processing****

* **Story 1:** Check for missing values, duplicates, and inconsistencies, and handle them appropriately.

* **Story 2:** Balance the dataset to ensure fair representation of all target classes.

* **Story 3:** Scale and normalize numerical features to improve model performance.

* **Story 4:** Split the dataset into training and testing sets for model development and evaluation.

****Epic 4: Model Building****

* **Story 1:** Train and evaluate a Decision Tree model on the prepared dataset.

* **Story 2:** Build and test a Random Forest model to improve predictive performance.

* **Story 3:** Implement a K-Nearest Neighbors (KNN) model and analyze its accuracy.

* **Story 4:** Train an XGBoost model and compare its performance with other models.

* **Story 5:** Evaluate all models using appropriate metrics and save the best-performing model for deployment.

****Epic 5: Application Building****

* **Story 1:** Design and develop HTML pages to create a user-friendly interface.

* **Story 2:** Build the Flask application and integrate the trained machine learning model.

* **Story 3:** Run, test, and validate the application to